

Benchmark #7

1 MeV H on Fe

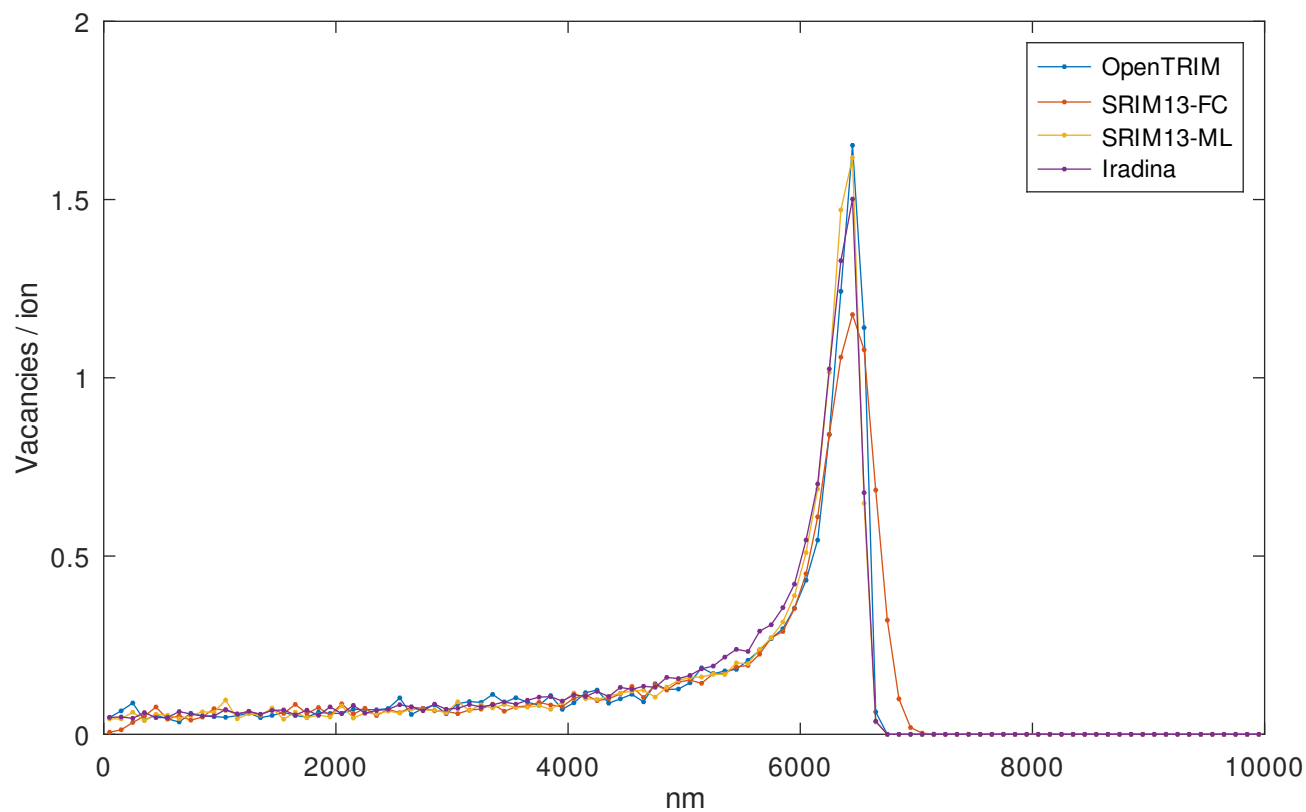
Ion energy E0 = 1e+06 eV

Target depth = 10000 nm

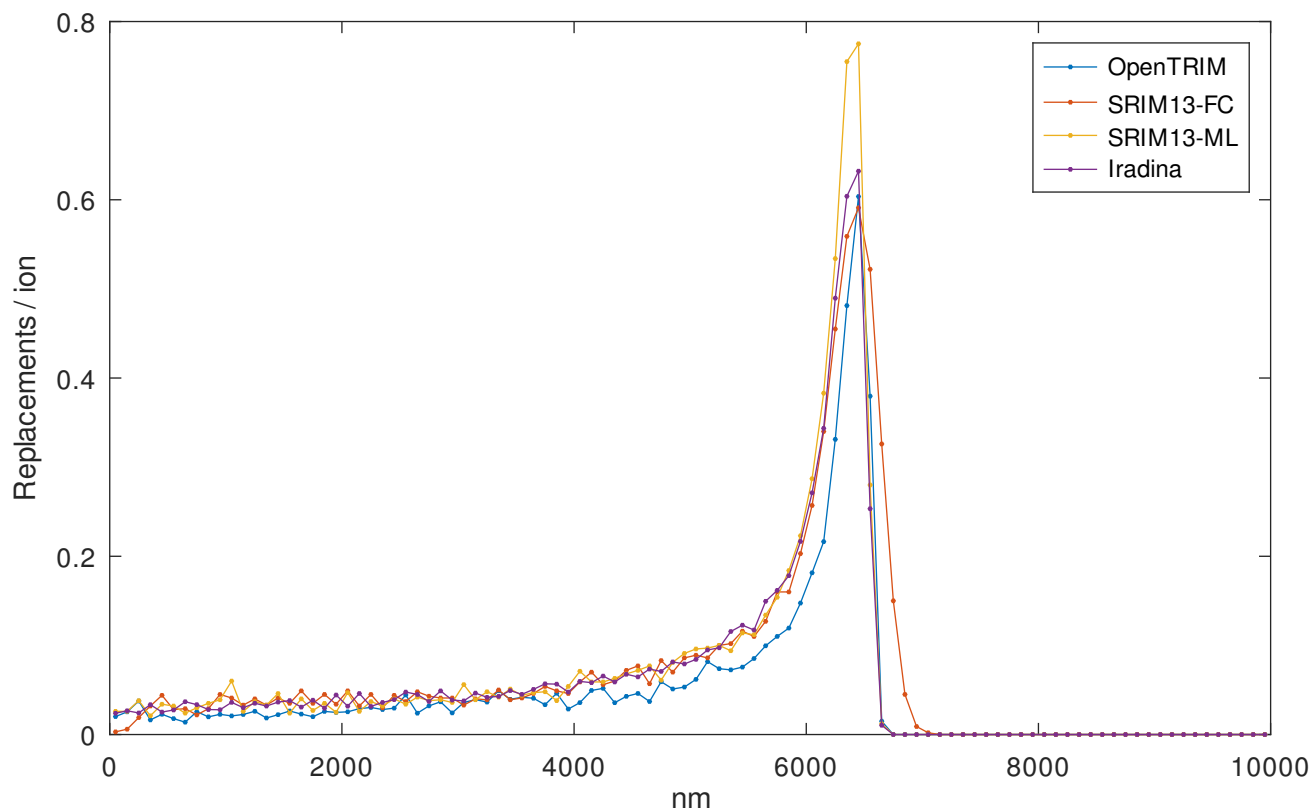
Summary Table

Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
V(Fe)	12	12.2	11.9	12.5
R(tot)	4.75	6.69	6.53	6.14
I(H)	1	1	1	1
$\Delta E_e(H)/E_0$	0.998	0.998	0.998	0
$\Delta E_e(r)/E_0$	0.000113	0.000104	0.000102	0
$\Delta E_e(tot)/E_0$	0.998	0.998	0.998	0.998
$\Delta E_{ph}(H)/E_0$	0.000752	0.000654	0.000915	0
$\Delta E_{ph}(r)/E_0$	0.000946	0.00154	0.00178	0
$\Delta E_{ph}(tot)/E_0$	0.0017	0.00219	0.00269	0.00183
$1 - (\Delta E_e + \Delta E_{ph})/E_0$	0	-0.00055	-0.000704	0

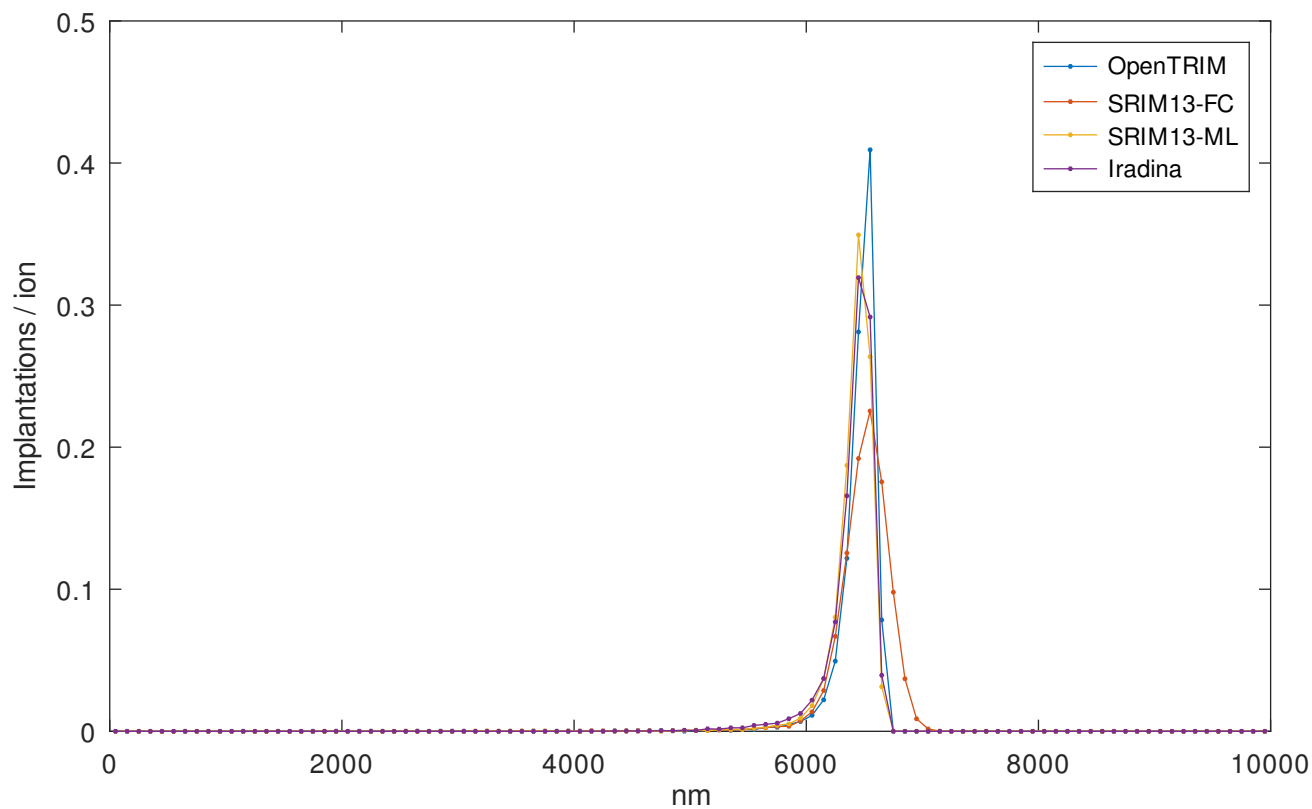
Vacancies of Fe in Fe



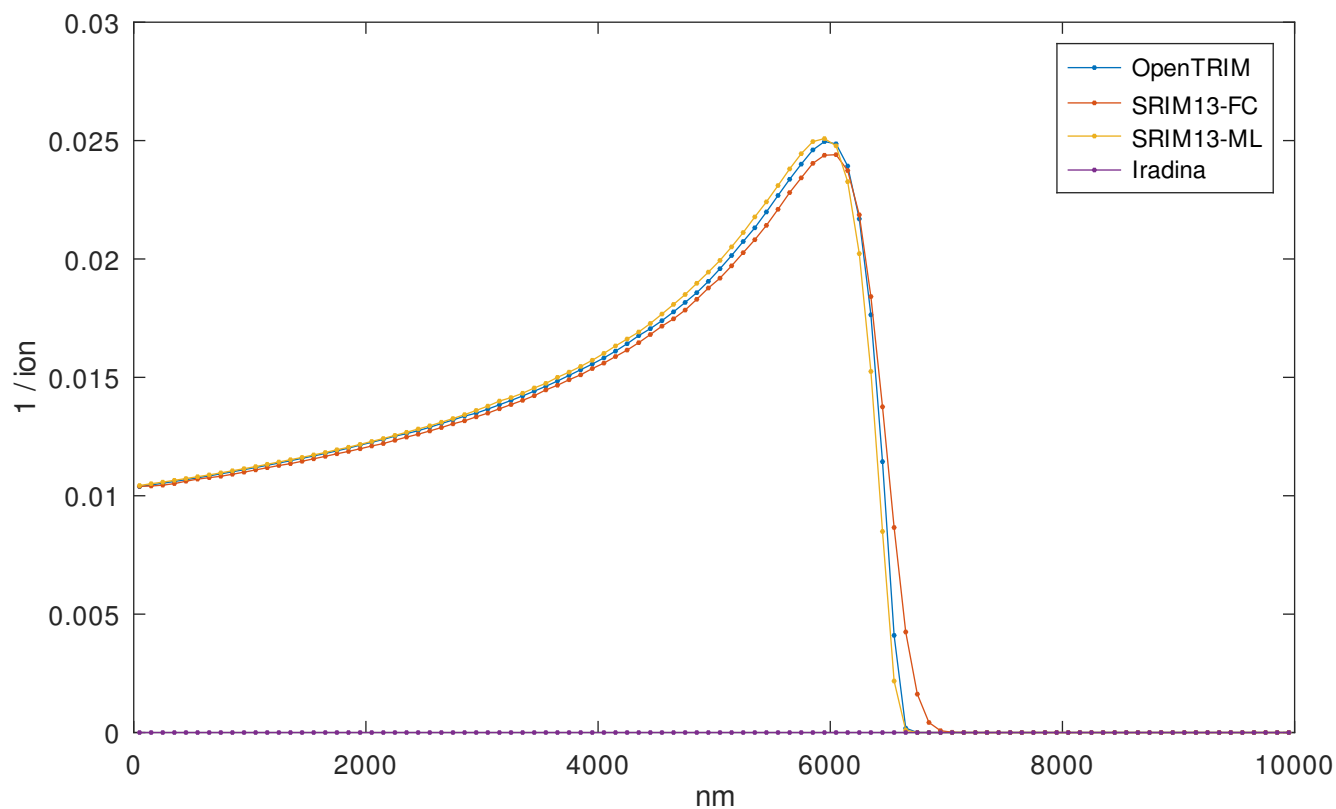
Replacements



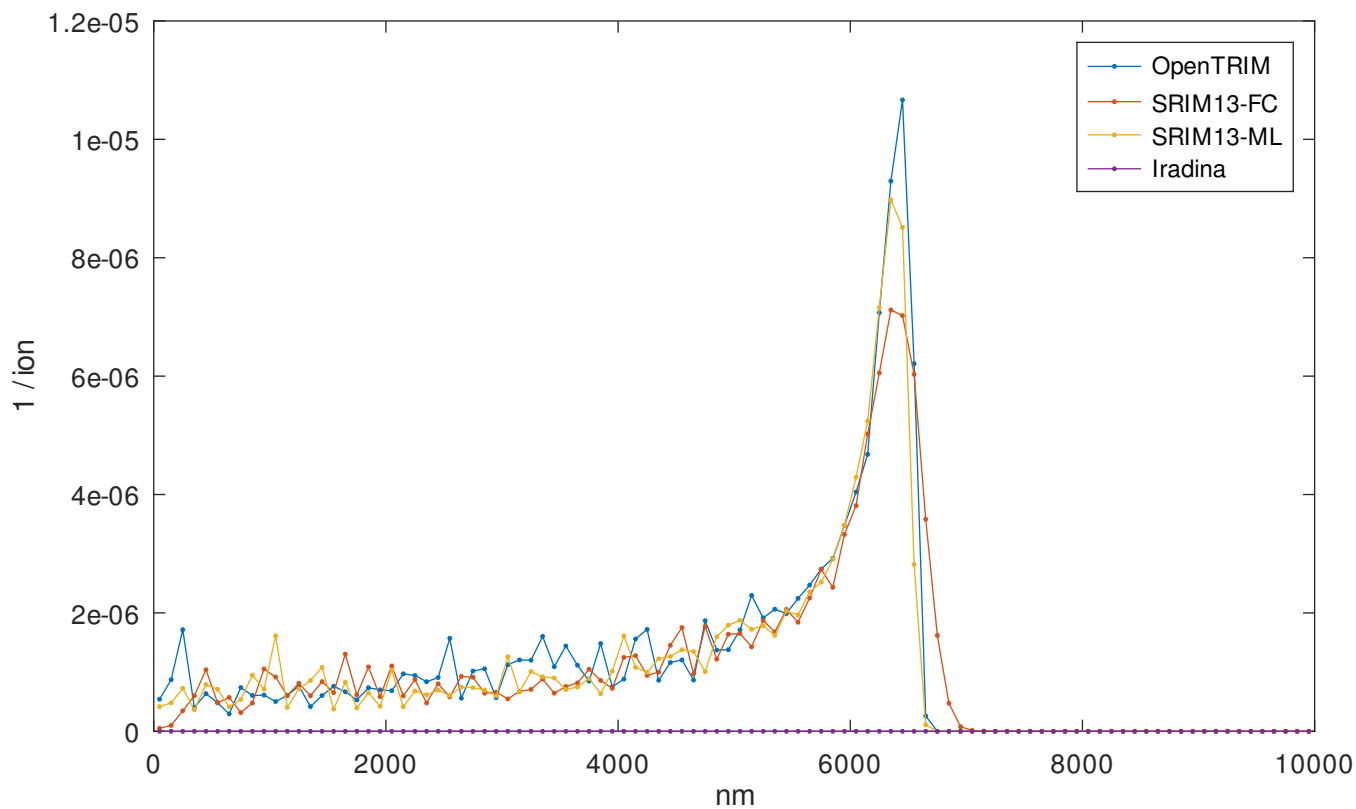
Implanted H ion



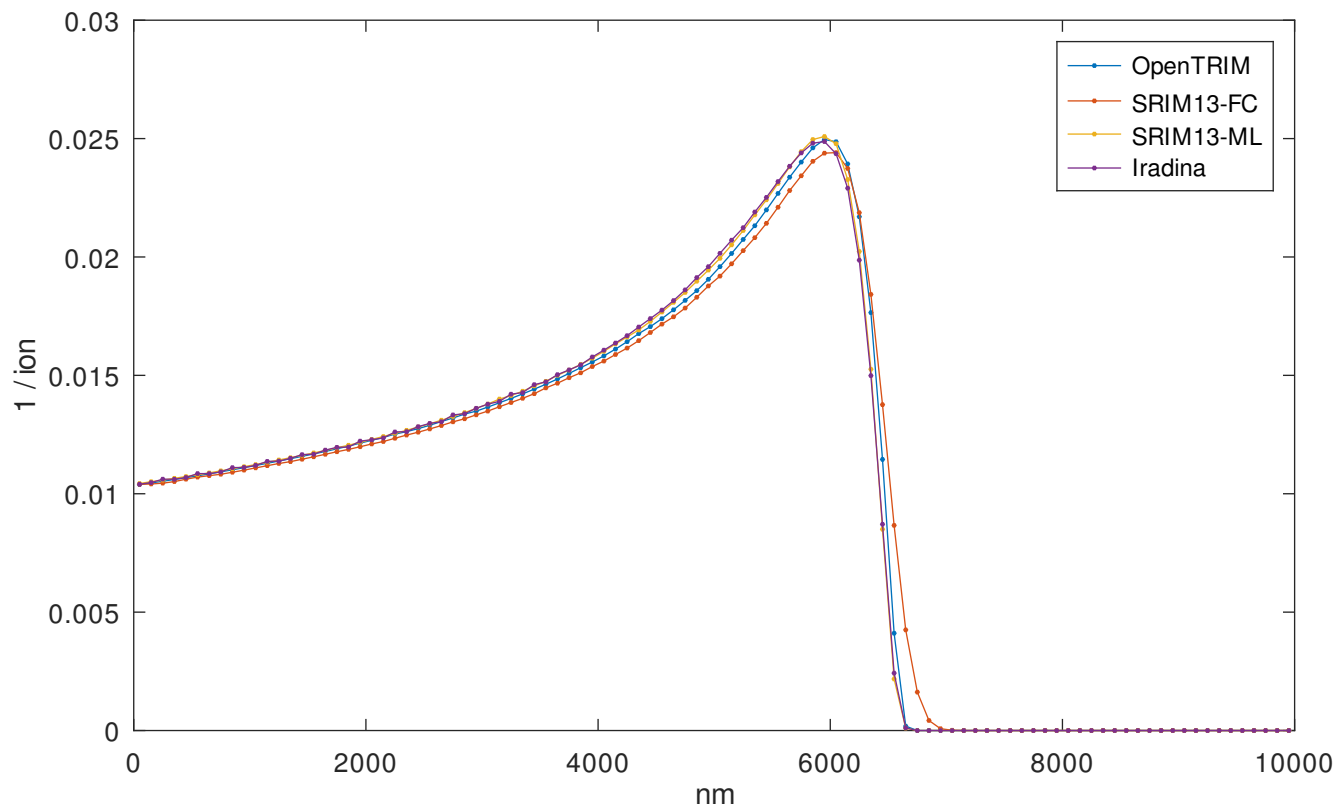
Electronic energy loss fraction $\Delta E_e/E_0$ by H ion



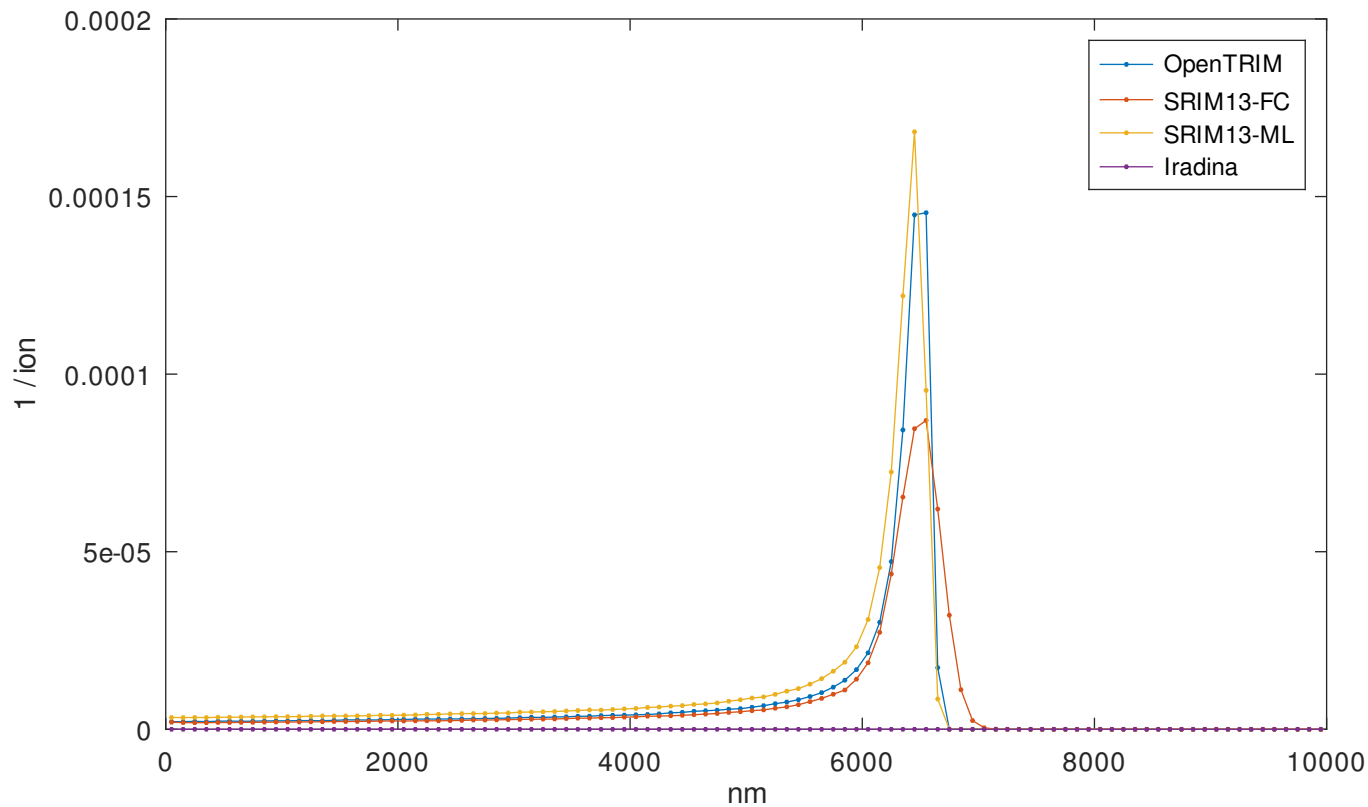
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



Total Electronic energy loss fraction $\Delta E_e/E_0$



Phonon energy fraction $\Delta E_{Ph}/E_0$ by H ion



Phonon energy fraction $\Delta E_{Ph}/E_0$ by recoils

